Project 2 EDA Report

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INTRODUCTION

This report presents an exploratory data analysis of a healthcare dataset containing 9,305 patient records from Lagos. The dataset includes patient demographics, vital signs, lab results, diagnosis, admission details and treatment outcomes.

The goal of this analysis is to explore the dataset structure, examine the distribution of key health variables and compare health indicators across diagnosis and outcomes. This report summarizes the steps taken and the key insights found.

DATA DESCRIPTION

This analysis uses the cleaned dataset produced in Project 1 which contains 9,305 patient records and 20 variables. It includes demographic information (Age, Gender, BMI), admission details (Admission Date, Admission Type), diagnosis information, vital signs (Systolic and Diastolic Blood Pressure), lab results (Glucose Level, Cholesterol, Hemoglobin), lifestyle factors (Smoking Status, Medication), hospital information (Doctor ID, Hospital), length of stay, readmission status and patient outcome.

All 9,305 records have complete data with no missing values. Numeric fields are stored as integers and floats while categorical fields such as Gender, Diagnosis, Admission Type and Outcome are stored as text.

METHODOLOGY

The analysis began by loading the clean dataset and examining its structure using shape (), info (), and describe () to understand the number of records, columns, data types and summary statistics. Value counts were generated for key categorical columns (Gender, Diagnosis, Admission Type) to understand the distribution of categories.

Outliers in BMI, Systolic BP, Diastolic BP, Cholesterol and Glucose Level were identified using the Interquartile Range (IQR) method, where values falling more than 1.5 times the IQR below the first quartile or above the third quartile were flagged.

A correlation matrix was computed for all numeric variables (Age, BMI, Systolic BP, Diastolic BP, Glucose Level, Cholesterol, Hemoglobin) using the Pearson correlation coefficient and visualized as a heatmap to identify any linear relationships between values.

Distributions of the key numeric variables were visualized using boxplot and histograms to examine their shape, spread and outliers.

Finally, the mean values of BMI, Systolic BP, Diastolic BP, Glucose Level and Cholesterol were compared across Diagnosis and outcome categories to identify any group level patterns. During this step, an inconsistency in the Outcome column's capitalization was identified and corrected using str.title() before re-running the comparison.

KEY FINDINGS

Distributions and Outliers

BMI, Systolic BP, Diastolic BP, Cholesterol and Glucose Level all show right skewed distributions with most values concentrated at lower end and a smaller number of high values forming a long tail. The IQR method identified outliers in each of these variables: 201 in Systolic BP, 83 in Diastolic BP, 104 in BMI, 132 in Cholesterol and 177 in Glucose Level. Due to the presence of skewness and outliers, the median provides a more robust measure of central tendency than the mean for these variables.

Correlation Analysis

The correlation matrix and heatmap show no strong relationships between the numerical health variables (Age, BMI, Systolic BP, Diastolic BP, Glucose Level, Cholesterol and Hemoglobin). No strong linear relationships were observed among the numerical variables which suggests that these variables vary independently of one another in this dataset.

Diagnosis and Cholesterol

Average values of BMI, blood pressure and glucose level were broadly similar across diagnosis but Cholesterol stood out. Artery Disease (196.7) and Heart Failure (197.9) had notably higher average Cholesterol than other diagnosis (mostly 172-180).

Outcome Comparison

After correcting inconsistent capitalization in the Outcome column, average health metrics were compared across Discharged, Improved, Transferred and Deceased outcomes. Differences were minimal across all variables, suggesting these health metrics alone do not strongly differentiate outcome groups.

CONCLUSION

The exploratory data analysis revealed that the dataset is complete and suitable for analysis. Several numerical variables exhibited right-skewed distributions and contained outliers. No strong linear relationships were observed among health indicators. Cholesterol levels were notably higher among patients diagnosed with Artery-Disease and Heart Failure. These findings provide useful insights for further statistical analysis and predictive modeling.

RECOMMENDATIONS

Given the high number of outliers in BMI, blood pressure, cholesterol and glucose further investigation is recommended to determine whether these represent genuine high-risk patients or data entry errors. The elevated cholesterol levels among Coronary-Artery-Disease and Heart Failure patients support continued cholesterol monitoring as part of cardiac care protocols. Since the numeric health variables show no strong correlations with each other. Finally, analysis should consider categorical relationships (e.g. Diagnosis, Admission Type) rather than relying on linear correlation alone.